

7(a)	(total) momentum before (decay) is zero or P has zero momentum	B1
	(total momentum after decay must be zero so) α -particle and Q have momenta in opposite directions (and therefore velocities are in opposite directions)	B1
7(b)	$p = 239 \text{ (u)} \times v \text{ or } 4 \text{ (u)} \times 1.6 \times 10^7$	C1
	$239 \text{ (u)} \times v = 4 \text{ (u)} \times 1.6 \times 10^7$	A1
	$v = 2.7 \times 10^5 \text{ m s}^{-1}$	
7(c)	$E_{(K)} = \frac{1}{2}mv^2$	C1
	$= \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times (1.6 \times 10^7)^2$	C1
	$= 8.5 \times 10^{-13} \text{ (J)}$	A1
	$= 8.5 \times 10^{-13} / 1.60 \times 10^{-13} \text{ (MeV)}$ $= 5.3 \text{ MeV}$	
7(d)(i)	1. R plotted at (95,147)	B1
	2. S plotted at (96,146)	B1
7(d)(ii)	(electron) <u>antineutrino</u>	B1